

# SAMANTHA LEE

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AI Native | Pipeline Development | Computer Graphics

## EDUCATION

University of Pennsylvania

*BSE in Computer Science (Digital Media Design)*

## SUMMARY

Senior Technical Artist building production tools and immersive experiences at Meta, Applied Intuition, and ILMxLAB. Specializes in pipeline development, Unreal Engine tooling, and real-time graphics, with hands-on experience integrating AI-based workflows to accelerate game and content development.

## EXPERIENCE

### Meta Reality Labs

Burlingame, CA

#### Senior Technical Artist

2026 - Present

- Developed AI-assisted game development workflows that empowered users to build games across platforms — investigated LLM-based 3D modeling, game design, and tool authoring skills
- Led a team to rapidly prototype different generative AI solutions for blocking out geometry to test gameplay, collaborating with game designers, artists, and engineers to optimize asset creation

#### Technical Artist

2025 - 2026

- Shipped switchable Home Vistas at Meta Connect 2025 — drove the fog transition tool with engineering, owned scene transition optimization, and reduced Quest 2 APK size by 30% via custom texture resolution settings
- Owned the lightprobe export tool powering dynamic avatar lighting across Horizon Home and Events; led content migration to Perforce and shipped a versioned toolchain that unblocked home environment delivery

#### Technical Artist (Contract)

2024 - 2025

- Built Unreal Engine C++ tooling for light baking, lightmap export, and asset processing — extending Meta's asset processor with custom logic to enable content ingestions to Meta Horizon Engine
- Drove tooling adoption with vendors and Meta artists, rapidly debugging asset processing, lightmap, and shader ingestion issues to unblock content delivery

### Applied Intuition

Sunnyvale, CA

#### Technical Artist

2023 - 2024

- Architected custom C++ Unreal Engine pipelines to generate diverse photoreal environments for real-world autonomous driving simulations
- Designed and maintained a full-stack in-product material editing system that powered synthetic data generation pipelines for training autonomous driving AI models
- Developed sensor simulation and procedural world generation tools using HLSL shaders and Houdini, enabling industry-leading synthetic data quality

## ILMxLAB

Remote

### Technical Artist Intern

2022

- Created VFX elements including procedural vine growth shaders and environmental effects for VR and AR productions
- Maintained a scene capture and projection pipeline to optimize real-time rendering performance
- Collaborated with lighting and environment artists to achieve desired looks through custom Unreal tooling

## Marvel Studios

Remote

### Production Technology Intern

2021

- Built a web application to centralize and streamline footage requests across 20+ productions
- Improved dev/deploy pipeline using Unix/Linux tooling to increase codebase scalability
- Partnered with Disney Animation engineers to adapt a shared application for Marvel production environments

## Apple

Remote

### GPU Tools Intern

2021

- Contributed to low-level graphics developer tools including the GPU Shader Debugger
- Collaborated with UI/UX teams to design intuitive interfaces for graphics development tools
- Built tooling supporting GPU-accelerated machine learning, VR, and AR technologies

## SKILLS

### LANGUAGES & FRAMEWORKS

C/C++ · Python · Java · JavaScript · TypeScript · React Native · HLSL · GLSL · OpenGL · USD · Unreal Blueprints

### TOOLS

Unreal Engine · Houdini · Maya · Photoshop · Git · Perforce · RenderDoc · Claude Code